

ay:

ix A

7x9

ce = 2

nk

s'7'

4] [7]
5] [8]
6] [9]

ndest?

$\mathbb{R}^3$

J.

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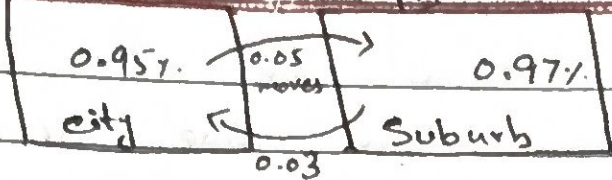
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dynamic system

$$\text{continuous} = \sum \text{discrete}$$

$$\frac{d}{dt} = \frac{d}{dt} - d_k$$

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so to predict the data
For K_{2025} we multiply them

$\Rightarrow$ Dynamical Systems.

$\Rightarrow$ The system in which change is occurring majorly.

e.g., Our atmosphere.

$\Rightarrow$ The model which is used to measure the dynamical system is

$$X_{k+1} = A X_k \quad \left\{ \begin{array}{l} \text{first order} \\ \text{difference equation} \end{array} \right\}$$

$$X_{k+1} - X_k = 0 \quad \left\{ \text{Difference Equation} \right\}$$

$\Rightarrow$ When we multiply M with X_k a time comes when $X_{k+1} = X_k$.

it means the change become constant/equilibrium which means

$$M \cdot X = X$$

$$MX - X = 0$$

$$(M-I)X = 0 \Rightarrow \boxed{AX=0}$$

$\Rightarrow$ It becomes a homogenous equation.

$\Rightarrow$ we have to find a matrix except for null to find the equilibrium.

so we get $X = \begin{bmatrix} 0.375 \\ 0.625 \end{bmatrix}$

This type of Matrix is called Stochastic Matrix.

$\Rightarrow$ Stochastic Matrix makes Markov Chain.

$\Rightarrow$ Markov Chain:-

Prob. vectors + Stochastic matrix.

if we have the data. For 2020 at 'X' we get equilibrium.

$$K_{2020} = \begin{bmatrix} 100 \\ 60 \end{bmatrix}$$

and we know the trend which is 'M'

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⇒ Eigen Vector: An eigen vector of $n \times n$ matrix A is a non zero vector $'X'$ such that $AX = \lambda X$ for some scalar $'\lambda'$.

Now check the eigen value for $A.v.$

$$= \begin{bmatrix} 3 \\ 15 \end{bmatrix} + \begin{bmatrix} -12 \\ -4 \end{bmatrix} = \begin{bmatrix} -9 \\ 11 \end{bmatrix}$$

⇒ It just shrinks and stretches but does not change its direction.

$$A.v = \begin{bmatrix} -9 \\ 11 \end{bmatrix} = \lambda \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

Eigen Value :- A scalar $'\lambda'$ is an eigen value of $'A'$ if there is a non-trivial solution $'X'$ of $AX = \lambda X$.

Question no:

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$$

Show that $'7'$ is the eigen value.

Example:

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}; u = \begin{bmatrix} 6 \\ -5 \end{bmatrix}; v = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$$AX = \lambda X.$$

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} X = 7X.$$

$$AX - 7X = 0$$

$$(A - 7I)X = 0.$$

Now argument it with $'0'$

$$\begin{bmatrix} 1 & 6 & | & 7 & 0 \\ 5 & 2 & | & 0 & 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1-7 & 6 & | & -6 & 6 \\ 5 & 2-7 & | & 5 & -5 \end{bmatrix}$$

Explore the null space of it and find the solution. if free variable is found it is eigen value otherwise not.
Eigen Space.

$$A.u = \begin{bmatrix} -24 \\ +20 \end{bmatrix} = \lambda u$$

$$\begin{bmatrix} -24 \\ 20 \end{bmatrix} = \lambda \begin{bmatrix} 6 \\ -5 \end{bmatrix}$$

(argument it with $'0'$)

eigen value = -4 .

found it is eigen value otherwise not.

$\Rightarrow$ we have at most distinct n values of eigen values.
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$$= \begin{bmatrix} -6 & 6 & 0 \\ 5 & 5 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 - x_2 = 0$$

$$0 = 0$$

$$x_1 = x_2$$

$$x_1 - x_2 = 0$$

$$x_1 = x_2$$

So eigen values are

$$\lambda = -4, 7$$

Question,

$$A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$$

Find all eigen values of 'A'

$$AX = \lambda X$$

$$AX - \lambda X = 0$$

$$(A - \lambda I)X = 0$$

So we do this

$$A - \lambda I$$

$$\Rightarrow \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix}$$

$\Rightarrow$ the 'X' is non-trivial vector.

So, it isn't invertible

$\Rightarrow$ So we have to find non-trivial vector

$\Rightarrow$ then find $\det(A - \lambda I) = 0$

$$(2-\lambda)(-6-\lambda) = 9 = 0$$

$$-12 - 2\lambda + 6\lambda + \lambda^2 - 9 = 0$$

$$\lambda^2 + 4\lambda - 21 = 0$$

$$\lambda^2 + 7\lambda - 3\lambda - 21 = 0$$

$$\lambda(\lambda+7) - 3(\lambda+7) = 0$$

$$(\lambda+7)(\lambda-3) = 0$$

$$\lambda = -7 \text{ or } \lambda = 3$$

$$\lambda = -7 \text{ or } \lambda = 3$$

$$\boxed{-7, 3} \text{ are}$$

the eigen values.

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$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; AX = \lambda X$$

What are the eigen vectors

of this

all - 2D vectors are eigen.

To check if a vector is eigen:

⇒ Multiply with the matrix.

⇒ if it is a multiple, then

it is a vector eigen.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x_2 - 3x_3 \\ x_2 \\ x_3 \end{bmatrix}$$

Question: Find eigen vector?

$$A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}; \lambda = 2$$

Sol:

$$AX = \lambda X$$

So we subtract

$$AX - \lambda X = 0$$

$$X(A - \lambda I) = 0$$

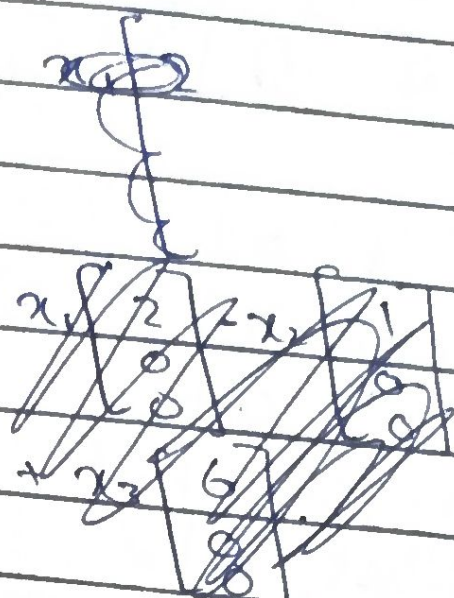
So,

$$= \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$2x_1 - x_2 + 6x_3 = 0$$



$$X = \begin{bmatrix} \frac{1}{2}x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} -3x_3 \\ 0 \\ x_3 \end{bmatrix}$$

$$X = x_2 \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

v_1 & v_2 are the 2 eigen vector corresponding to the eigen value $\lambda = 2$ which are extracted from null-space

$AX = \lambda X$ is true

For all vectors in the span = $\left\{ \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$(A - \lambda I)X = 0$$

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Q. what it means for a matrix to have $\lambda = 0$?

$$Ax = 0$$

Sol: If one of the eigen value is '0' what it means....?

$\Rightarrow$ we are finding non-trivial solution but at '0' it is a trivial and if det is zero it is singular.

which means pivots are less and it is not invertible

AND If we have '0' eigen value, it is not-invertible.

For singular matrix

$$\det(A) = 0.$$

which is

$$\det(A - \lambda I) = 0.$$

$\Rightarrow$ what to do if eigen value is unknown....?

Sol: In that case we subtract λ from diagonal values.

then take its determinant

$\Rightarrow$ after that we get a n^{th} degree polynomial with variable λ .
unknown

so we get value of the eigen values.

and that equation is called characteristic equation.

Question.

$$A = \begin{bmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ 0 & a_{22}-\lambda & a_{23} \\ 0 & 0 & a_{33}-\lambda \end{bmatrix}$$

$$\det A = \begin{cases} (-1)^r \times \text{Product of non-zero diagonal} \\ 0 \end{cases}$$

• diagonal entries (on the main diagonal) in echelon form
 r = row interchanges count.

Q. Find det. of A

$$A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 4-\lambda & -1 & 6 \\ 2 & 1-\lambda & 6 \\ 2 & -1 & 6-\lambda \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1/4 & 3/2 \\ 0 & 3/2 & 9/2 \\ 0 & -2 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1/4 & 3/2 \\ 0 & 2 & -2 \\ 0 & -1/2 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1/4 & 3/2 \\ 0 & 2 & -2 \\ 0 & 0 & 9/2 \end{bmatrix}$$

$$\begin{aligned} &= \frac{-1}{4} \\ &= \frac{1}{4} + 1 \\ &= \frac{1+4}{4} = \frac{5}{4} \\ &= \frac{5}{4} \times \frac{9}{2} = \frac{45}{8} \end{aligned}$$

$$\begin{aligned} &= -1 + 1 \\ &= \frac{-1+1}{4} = 0 \end{aligned}$$

$$\begin{bmatrix} 4 & -1 & 6 \\ 0 & 2 & -2 \\ 0 & 0 & 9/2 \end{bmatrix} \Rightarrow 4(8+6) + 1(16-1) + 6(-2-2) = 36 \text{ Ans.}$$

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Find det of

$$A = \begin{bmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ 0 & a_{22}-\lambda & a_{23} \\ 0 & 0 & a_{33}-\lambda \end{bmatrix}$$

$$B - \lambda I = P^{-1}(A - \lambda I)P$$

$$\det(B - \lambda I) = \det(P^{-1}(A - \lambda I)P)$$

$$= \det P^{-1} \cdot \det(A - \lambda I) \cdot \det P$$

$$= (a_{11}-\lambda)(a_{22}-\lambda)(a_{33}-\lambda) - 0$$

$$= a_{11}-\lambda(a_{22}-\lambda \cdot a_{33}-\lambda-0) - 0$$

$$\begin{bmatrix} 0 & 0 \end{bmatrix} + a_{13} \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$\det(B - \lambda I) = \det P^{-1} \cdot \det P \cdot \det(A - \lambda I)$$

$$= \det(P^{-1} \cdot P) \cdot \det(A - \lambda I)$$

$$\det(B - \lambda I) = \det(A - \lambda I)$$

$$= (a_{11}-\lambda)(a_{22}-\lambda)(a_{33}-\lambda) = 0$$

Now find eigen values

$$a_{11}-\lambda=0 ; a_{22}-\lambda=0 ; a_{33}-\lambda=0$$

$$a_{11}=\lambda ; a_{22}=\lambda ; a_{33}=\lambda$$

$\Rightarrow$ For a matrix 'A' to be diagonalized, it has to be similar to a diagonal matrix.

Similarity of Matrix

$$A = P D P^{-1}$$

$\hookrightarrow$ Diagonal matrix

'A' & 'B' are two similar

if there exist a matrix 'P' which is invertible

$\Rightarrow$ It only scales the matrix as the values are in diagonal.

$$P^{-1} A P = B$$

$\Rightarrow$ If P is made up of eigen vectors it is easy to solve

$\Rightarrow$ If A is order nxn P must be of nxn.

$\Rightarrow$ If the A has n number of eigen vectors

$\Rightarrow$ In case of similar matrix both also have same characteristic expression.

$\Rightarrow$ eigen values also same

Now,

$$B = P^{-1} A P$$

$$B - \lambda I = P^{-1} A P - \lambda I$$

$$= P^{-1} A P - \lambda P^{-1} P$$

$$= P^{-1} (A P - \lambda P)$$

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Find determinant
using this.

$$\det(A) = \begin{cases} (-1)^r \cdot \text{Product of diagonal entries} \\ \text{in echelon form} \\ 0 \end{cases}$$

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Sol:

$$\begin{array}{l} \text{r1} \\ \text{r2} \end{array} \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$\text{r2} \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & -6 & -1 \end{bmatrix}$$

$$\begin{array}{l} \text{r3} \\ \text{r2} + 3\text{r1} \end{array} \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$= (-1)^3 \cdot (-1)(2)(1)$$

$$= (-1)(2)$$

$$= -2 \text{ Ans.}$$

Question Find
eigen value

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

in this matrix.

these are the two
eigen value = 3, 2.

$$AX = \lambda X$$

$$AX - \lambda X = 0$$

$$X(A - \lambda I) = 0$$

$$\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \text{ After this}$$

$$\begin{array}{l} \text{r1} \\ \text{r2} \end{array} \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$$

find the null space

$$\text{so } \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ by } \lambda = 3$$

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* Both are same.

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Now $\lambda = 2$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= x_1 + x_2 = 0$$

$$x_1 = -x_2$$

So $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ or $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ are.

=> If P is made up of eigen vectors it becomes easier to solve

=> If we have a diagonal matrix the diagonal entries are eigen values

Q: $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}; B = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$
eigen value of A & B = 2.

Ans: They have same eigen values but they are not similar. but we need to find more proof of similarity.

$$Ax = \lambda x$$

$$x(A - \lambda I)$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

i) $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ It's not true.

dimension (null A) = 2

$$B - \lambda I$$

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

ii) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ dimension = 1

So not similar.

=> AND To check similarity of dimension similar = eigen value & characteristic equation

* => If column space's dimension is also '0'
* dimension of null space is also '0'

$$\Rightarrow Ax = \lambda x$$

if we square A so,
 $A^2 x = \lambda^2 x$

=> If we want to get power of a matrix and first check if it is diagonalised able so, $A; A^2; \dots$
 $A.A = P D P^{-1} \cdot P D P^{-1}$
 $= P D^2 P^{-1}$

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multiply the diagonal is difficult.
so we just square the
diagonal entries

=> To convert any matrix
squared into diagonal
there must be a matrix
which is invertible so.

$$A = P D D^{-1}$$

9-Dec-2025.

$$A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\lambda_1 = 1; \lambda_2 = -2$$

Question: Check if the
matrix is diagonalized?
(3rd eigen is repeated?)

=> If eigen values are distinct
the matrix becomes diagonalized.

=> If eigen values are not distinct
it might or might not be
diagonalized.

=> To find, we find $A - \lambda I$
for each eigen value, then
row reduce it, and then
we get vectors and
the count of vectors
is equal to 'n'

For $\lambda = 1$

$$A - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ -4 & -5 & -3 \\ 3 & 3 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ -4 & -5 & -3 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ 0 & -9 & -3 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ 0 & 3 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ -3 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 3 \\ -3 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ -3 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ 1 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ 1 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

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$$= \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ 0 & -3 & -3 \\ 0 & 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x_1 + 4x_2 + 3x_3$$

$$x_2 + x_3$$

$$x_3$$

$$\begin{bmatrix} x_1 + 4x_2 + 3x_3 \\ x_2 + x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{For } \lambda_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}; \lambda_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

It is not distinct and

$$\left[\begin{array}{ccc|c} 3 & -3 & 0 & 0 \\ 2 & -2 & 0 & 0 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 3 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\text{Example } A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$$

Sol:

Find A^8 ?

$$A = P D P^{-1}$$

$$\text{So, } A^8 = P D^8 P^{-1}$$

Find A^8 : $\Rightarrow$ Check if it is diagonalized able

Find eigen values

$$\det(A - \lambda I) = \det \begin{bmatrix} 4-\lambda & -3 \\ 2 & -1-\lambda \end{bmatrix} = 0$$

$$(4-\lambda)(-1-\lambda) + 6 = 0$$

$$-4 - 4\lambda + \lambda + \lambda^2 + 6 = 0$$

$$\lambda^2 + 3\lambda + 2 = 0$$

$$\lambda^2 - 2\lambda - \lambda + 2 = 0$$

$$\lambda(\lambda-2) - 1(\lambda-2) = 0$$

$$(\lambda-2)(\lambda-1) = 0$$

$$\lambda = 2; \lambda = 1;$$

 $\Rightarrow$ Values are distinct
it is diagonalized
able.

Now, diagonalize it

$$A - I = \begin{bmatrix} 3 & -3 & 0 \\ 2 & -2 & 0 \end{bmatrix}$$

$$\lambda_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} =$$

$$\begin{bmatrix} 3 & -3 & 0 \\ 2 & -2 & 0 \end{bmatrix}$$

$$\Rightarrow 3x_1 = 3x_2$$

$$x_1 = x_2$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

v_2 is dependent on v_1

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and for

$$\lambda = 2$$

$$\begin{bmatrix} 4-2 & -3 \\ 2 & -1-2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -3 & 0 \\ 2 & -3 & 0 \end{bmatrix}$$

$$\begin{aligned} 2x_1 &= 3x_2 \\ x_1 &= \frac{3}{2}x_2 \end{aligned}$$

$$\begin{bmatrix} 2 & -3 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3/2 \\ 0 & 0 \end{bmatrix}$$

$$2x_1 = 3x_2$$

$$x_1 = \frac{3}{2}x_2$$

$$x_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Now Find A^8

$$A^8 = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$A^8 = \begin{bmatrix} 256 & 0 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow$ If eigen value is '0' it is not diagonalized because it is trivial and we have to find a free variable and non-consistent.

Question no 3:

$$M = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$$

$$x_0 = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

$\Rightarrow$ If we have probability vector this stage becomes a chain

$$X_{k+1} = AX_k$$

$k=0,1,2,3,4,\dots$

Find equilibrium vector?

$$X_1 = AX_0$$

$$X_2 = AX_1 = A^2X_0$$

$$X_3 = AX_2 = A^3X_0$$

$$X_3 = AX_2 = A^3X_0$$

what is the driving force?

Ans: It is the eigen value

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So now we find eigen values

$$M - \lambda I = \begin{bmatrix} 0.95 - \lambda & 0.03 \\ 0.05 & 0.97 - \lambda \end{bmatrix}$$

Now find

$$\det(M - \lambda I) = (0.95 - \lambda)(0.97 - \lambda) - 0.0015$$

$$0.9215 - 0.95\lambda - 0.97\lambda + \lambda^2 - 0.0015$$

$$\lambda^2 - 1.92\lambda + 0.92 = 0$$

$$\lambda^2 - 1.92\lambda + 0.92 = 0$$

$$\lambda^2 - \lambda - 0.92\lambda + 0.92 = 0$$

$$\lambda(\lambda - 1) - 0.92(\lambda - 1) = 0$$

$$(\lambda - 1)(\lambda - 0.92) = 0$$

$$\lambda = 1 \quad ; \quad \lambda = 0.92$$

$$M - \lambda I = \begin{bmatrix} -0.05 & 0.03 \\ 0.05 & -0.03 \end{bmatrix}$$

Multiply by 100.

$$= \begin{bmatrix} -5 & 3 & 0 \\ 5 & -3 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - \frac{3}{5}x_2 \\ x_2 = x_2 \end{bmatrix}$$

$$= x_2 \begin{bmatrix} 3/5 \\ 1 \end{bmatrix}$$

To remove fraction do multiply $x_2 = 5$
 so $v_1 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

For $\lambda_1 = v_1 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

For $\lambda_2 = v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

So,

$$P_B = \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}$$

and from this we can calculate

$$x_0 = c_1 v_1 + c_2 v_2$$

How to find c?

$\Rightarrow$ we are given x_0 .

$$x = P \begin{bmatrix} x \end{bmatrix}_B$$

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} x \end{bmatrix}_B = P_B^{-1} x$$

So,

~~As we are given~~

As we are given $x_1 = A x_0$

$$x_1 = A(c_1 v_1 + c_2 v_2)$$

$$x_1 = A c_1 v_1 + A c_2 v_2$$

$$\because A v_1 = \lambda v_1$$

So,

$$x_1 = \lambda_1 c_1 v_1 + \lambda_2 c_2 v_2$$

$$x_1 = c_1 v_1 + c_2 v_2 (0.92)$$

$$x_2 = A x_1 = A(c_1 v_1 + c_2 v_2 (0.92))$$

$$x_2 = c_1 v_1 + c_2 v_2 (0.92^2)$$

So, for k^{th} value

$$x_k = c_1 v_1 + c_2 v_2 (0.92)^k$$